

The Aristotelian Principle

A Universal Equilibrium Condition
from Atomic to Cosmological Scales

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<https://github.com/hmahaffeyges/IAM-Validation>

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Abstract

The virial theorem, established by Clausius in 1870, states that for any system bound by a $1/r$ potential, the time-averaged kinetic and potential energies satisfy $2\langle K \rangle + \langle V \rangle = 0$. This result has been verified from atomic scales to galaxy clusters across 37 orders of magnitude in physical scale.

We report evidence that the virial theorem is the special case of a deeper principle operating across the full dynamic range of physics — from the hydrogen atom to the ratio of the universe’s accumulated dark energy to its total vacuum energy. We propose the name **the Aristotelian Principle** for this general law: *the equilibrium condition between unrealized potential and accumulated actuality that governs the structure of any physical system at every scale where stable existence is possible.*

Within the Informational Actualization Model (IAM) [Mahaffey, 2026a], the Aristotelian Principle is identified as the physical foundation from which the coupling constant $\beta_m = \Omega_m/2$, the sector split $\mu < 1$, $\Sigma = 1$, the M – σ relation, the Bekenstein-Hawking entropy partition, the Koide lepton mass relation, the cosmological virialisation $\Omega_m/[\beta_m E(a=1)] = 2$ exactly, and the cosmological constant ratio $\Lambda/\rho_{\text{vac}} = (l_P/l_H)^2 \times (\Omega_b/\Omega_m)$ all derive. Tables 1 and 2 present 22 expressions of the Aristotelian Principle across 33 papers and every physical domain in which the IAM framework has been tested.

A critical structural observation accompanies this inventory: three of the four fundamental forces — gravity, electromagnetism, and the strong nuclear force — satisfy the Aristotelian Principle exactly within their respective domains. The weak nuclear force is the sole structural exception, excluded not by coincidence but by physical necessity: the force whose function is to enable irreversible transitions

between stable states cannot satisfy an equilibrium condition. This exception is itself evidence for the principle’s physical content, distinguishing forces of stable structure from the force of irreversible change.

The virial theorem retains its name and its place in the literature as the special case for bound systems with $1/r$ potentials. The Aristotelian Principle names the broader law of which the virial theorem is one expression, proposed in honour of the person who first identified the underlying equilibrium condition between potential and actual as the fundamental condition of finite existence.

Keywords: Aristotelian Principle; virial theorem; gravitational decoherence; cosmological constant; dark energy; dark matter; weak nuclear force; Informational Actualization Model; universal equilibrium

1 Introduction: Beyond the Virial Theorem

The virial theorem has governed the energy partitioning of physical systems since Clausius stated it mathematically in 1870 [Clausius, 1870]. For any system bound by a potential $V \propto r^{-n}$, the time-averaged kinetic and potential energies satisfy:

$$2\langle K \rangle + n\langle V \rangle = 0 \tag{1}$$

For the $1/r$ potential ($n = 1$) — governing both Newtonian gravity and the Coulomb interaction — this gives $\langle K \rangle = \frac{1}{2}|\langle V \rangle|$. The 1/2 partition has been confirmed from the hydrogen atom (10^{-10} m) to galaxy clusters (10^{23} m) and to the cosmological coupling constant of IAM confirmed by 17 Planck 2018 MCMC chains [Mahaffey, 2026b].

A legitimate question arises when the coupling constant derivation is extended: why should the virial theorem apply at cosmological scales? The virial theorem in its standard form describes bound systems that have reached equilibrium. The cosmological application in IAM — deriving $\beta_m = \Omega_m/2$ from the virial partition of gravitational decoherence energy — extends the theorem beyond its original domain.

This paper argues that the question rests on a category error. The framework presented here is not an application of the virial theorem at cosmological scales. It is the identification of a deeper principle — the **Aristotelian Principle** — of which the virial theorem is the special case for bound systems with $1/r$ potentials. The cosmological coupling constant, the dark sector partition, the cosmological constant ratio, and every other result discussed below are expressions of this deeper principle at different scales. None of them are extensions of the virial theorem. All of them are special cases of the Aristotelian Principle.

The Aristotelian Principle states: *in any physical system where stable existence is possible, the equilibrium between unrealized potential and accumulated actuality is governed by*

the same fundamental condition, with the $1/2$ partition appearing wherever $1/r$ potentials dominate in three-dimensional space, and more general expressions of the potential-actual ratio appearing at other scales and in other physical contexts.

Section 2 presents the complete evidence inventory across 33 papers and every physical domain where IAM has been tested. Section 3 analyzes the four fundamental forces and the structural exception of the weak nuclear force. Section 4 presents the MCMC confirmation as the strongest empirical test of the principle. Section 5 presents the cosmological constant as the Aristotelian Principle operating at the ratio of the universe's accumulated actuality to its total potential. Section 6 states the principle precisely. Section 7 presents the petition for naming. Section 8 concludes with an observation about the principle's place in the history of natural philosophy.

2 The Complete Evidence Inventory

Tables 1 and 2 present every expression of the Aristotelian Principle identified across the IAM framework. The table is organized by physical domain and scale, from atomic (electromagnetic) through stellar, cluster, galactic, cosmological, and particle physics scales, culminating in the new cosmological constant result. The weak nuclear force exception is highlighted separately.

The table documents 22 distinct expressions of the principle across 33 papers and every physical scale at which physics has been tested. The range extends from the hydrogen atom at 10^{-10} m to the ratio of the Planck scale to the Hubble scale — 10^{61} orders of magnitude in length, 10^{122} orders of magnitude in energy density.

Table 1: The Aristotelian Principle — Part A: Core virial theorem domain. The 1/2 partition verified across 37 orders of magnitude in physical scale. **Blue rows**: confirmed by 17 independent Planck 2018 MCMC chains.

Domain	Scale / Force	A.P. Ratio	Result	Precision
Atoms (H–Xe, 20 elements)	10^{-10} m, EM	$T/ V $	1.0000000000	Machine precision [Mahaffey, 2026b]
Molecules (H ₂ –CO ₂ , 10)	10^{-9} m, EM	$T/ V $	1.0000000000	Exact at equil. geom. [Mahaffey, 2026b]
Equipartition theorem	10^{-9} – 10^{-2} m, Thermal	$kT/2$ per DOF	1/2 exact	Exact in limit [Mahaffey, 2026b]
Solar structure	10^9 m, Gravity	$K/ U $	≈ 0.5	$\sim 10\%$ [Mahaffey, 2026b]
White dwarfs (Chandrasekhar)	10^7 m, Gravity	M_{Ch}	$1.44 M_{\odot}$	Exact, observed [Mahaffey, 2026b]
Galaxy clusters (~ 20)	10^{23} m, Gravity	$K/ U $	≈ 0.5	$\sim 20\%$ [Mahaffey, 2026b]
Virial efficiency $\langle \eta_{\text{vir}} \rangle$ (6 N-body)	10^{22} – 10^{25} m, Gravity	$2K/ U $	0.815 ± 0.025	6 independent groups [Mahaffey, 2026c]
Cosmological coupling β_m	10^{26} m, Gravity	β_m/Ω_m	0.1583 ± 0.0033 (post.)	0.2σ , 17 chains [Mahaffey, 2026a]
Scale range	10^{-10} m (H atom) to 10^{26} m (cosmic horizon) — 37 orders of magnitude			

Table 2: The Aristotelian Principle — Part B: IAM extended domain. Expressions of the principle beyond the virial theorem’s original domain. **Green rows**: new results from this work (March 2026). **Red row**: weak nuclear force — structural exception that confirms the principle’s physical content.

Domain	Scale	A.P. Expression	Result / Precision
H_0 sector split	Cosmological	$H_0^{\text{matter}}/H_0^{\text{photon}}/\sqrt{1+\beta_m}$	$= 72.26/67.16 \text{ km/s/Mpc}$, both $< 1\sigma$ [Mahaffey, 2026a]
σ_8 suppression	Cosmological	$\sigma_8^{\text{IAM}} = 0.800$	0.7998 ± 0.0058 , 0.1σ [Mahaffey, 2026h]
DESI transition red-shift	$z \approx 0.7$	$z_t^{\text{IAM}} = 0.718$	Inside all $H(z)$ compilations [Mahaffey, 2026g]
M – σ relation (SMBHs)	10^{20} m	$M_{\text{BH}} \propto \sigma^4$, $f_{\text{geom}} = \Omega_m/(2\pi)$	2% from observed, 0.7σ (115 galaxies) [Mahaffey, 2026i]
Lensing-dynamics discrepancy	10^{23} m	$M_{\text{lens}}/M_{\text{dyn}} = 1/\mu(z)$	Consistent with cluster data [Mahaffey, 2026j]
BH entropy partition	$r_s = 2GM/c^2$	$E_{\text{Landauer}} (\ln 2/2) M_{\text{BH}} c^2$	$= 34.7\%$ of rest mass (universal) [Mahaffey, 2026k]
Sector split (measurement problem)	$\Delta\tau = 0$, null	$\Sigma = 1$ (null), $\mu < 1$ (time-like)	Exact from metric signature [Mahaffey, 2026l]
Electron rest mass (conditional)	λ_C	$m_e c^2 = E_{\text{bit}} \times N(m_e)/\alpha^{5/2}$	Conditional derivation [Mahaffey, 2026n]
Koide relation (conditional)	Lepton scale	$Q = 2/3$ from Z3 virial on S^2	Conditional on holographic ansatz [Mahaffey, 2026o]
Dark sector virial partners	$a = 1$	$\Omega_m/[\beta_m E(1)] = 2$ exactly	Coincidence problem dissolved [Mahaffey, 2026d]
Cosmological constant ratio	l_P to l_H	$\Lambda/\rho_{\text{vac}} = (l_P/l_H)^2 \times (\Omega_b/\Omega_m)$	2.17×10^{-123} (obs: 1.14×10^{-123}), factor 1.9 [Mahaffey, 2026f]
Force separation sequence	10^{-15} – 10^{26} m	Each separation enables next virial scale	Protons $\rightarrow$ Atoms $\rightarrow$ Halos $\rightarrow$ Cosmos [Mahaffey, 2026e]
Weak nuclear force (exception)	10^{-18} m	Structurally excluded: force of becoming	Exclusion confirms the principle [Mahaffey, 2026e]

3 The Four Fundamental Forces and the Structural Exception

The Aristotelian Principle is satisfied by every force that maintains stable structures. Three of the four fundamental forces fall into this category.

Gravity operates through a $1/r$ potential in three-dimensional space. Gauss’s law requires the $1/r^2$ force law from any point source in 3D, giving the $1/r$ potential and therefore the exact virial ratio of $1/2$ for any gravitationally bound system. The Aristotelian Principle is verified from stellar structure to galaxy clusters to the cosmological coupling constant at 0.2σ from 17 independent MCMC chains.

Electromagnetism operates through the same $1/r$ Coulomb potential. The Aristotelian Principle is verified at machine precision for 20 atoms and 10 molecules from published Hartree-Fock energies [Mahaffey, 2026b]. The mean virial ratio $\eta = -T/E = 1.0000000000$ with standard deviation zero. This is not an approximation — for converged Hartree-Fock, the virial ratio must equal unity exactly as a mathematical consequence of the $1/r$ potential.

The strong nuclear force produces the first virialized structures in cosmic history: protons forming at $t \approx 10^{-6}$ seconds when quark-gluon plasma underwent QCD confinement. The proton has not been observed to decay in 10^{33+} years. The Chandrasekhar white dwarf limit — defined by the virial equilibrium between electron degeneracy pressure and gravitational collapse — is an exact observational consequence of the Aristotelian Principle, confirmed at $M_{\text{Ch}} = 1.44 M_{\odot}$ [Chandrasekhar, 1931].

The weak nuclear force is the structural exception. It violates parity [Wu et al., 1957], violates CP symmetry [Cronin & Fitch, 1964], and is responsible for beta decay, stellar fusion, and the matter-antimatter asymmetry that produced the observable universe [Sakharov, 1967]. It cannot satisfy the Aristotelian Principle not because it fails to do so but because satisfying it would prevent the weak force from fulfilling its function.

The Aristotelian Principle describes the equilibrium of stable structures. The weak force’s function is to enable irreversible transitions between stable structures — to dissolve one equilibrium state and create another. A force in virial equilibrium maintains what is. The weak force by design transforms what is into what comes next. These functions are structurally incompatible.

The exception is itself evidence. A principle that holds for all forces of stable structure and is violated by precisely the one force whose function is structural instability is not a numerical coincidence. It is a statement about the physical character of forces. The Aristotelian Principle distinguishes forces of being from the force of becoming. That distinction has physical content and observable consequences — including the fact that CP violation in the weak sector set Ω_b at electroweak symmetry breaking, contributing 15.6% of β_m , while the remaining 84.4% accumulated through gravitational decoherence

governed by the Aristotelian Principle over 13.8 billion years [Mahaffey, 2026e].

4 The MCMC Confirmation: The Strongest Empirical Test

The most rigorous empirical test of the Aristotelian Principle is the Planck 2018 MCMC posterior returning $\beta_m = 0.1583 \pm 0.0033$ against the theoretically predicted $\beta_m = \Omega_m/2 = 0.1577$ — a confirmation at 0.2σ — from 17 independent converged chains with no free parameters, no discarded runs, and no excluded redshift ranges [Mahaffey, 2026a].

The significance of this confirmation is best understood through what the MCMC sampler does not know. The sampler has no knowledge of the Aristotelian Principle, the virial theorem, or the physical argument that $\beta_m = \Omega_m/2$. It samples the posterior probability distribution of the cosmological parameters given the Planck likelihood. The posterior peak at $\beta_m = 0.1583$ is not guided toward the Aristotelian value — it arrives there because the Planck data is consistent with that value and inconsistent with significantly different values.

The Aristotelian Principle was derived from the virial theorem before a single chain ran. The posterior confirmed it. This sequence — prediction before confirmation, from an independent measurement that had no knowledge of the prediction — is the gold standard of scientific evidence.

Table 3 summarises the MCMC confirmation results.

Table 3: MCMC confirmation of the Aristotelian Principle coupling constant. All 17 chains converged with $R-1 < 0.01$. No chains discarded. Zero free parameters beyond CDM.

Chain set		Dataset	β_m posterior vs A.P. prediction (0.1577)
Level 1	(12 chains)	Planck 2018 + combinations	0.1577–0.1592 ($< 0.3\%$ from A.P.)
Level 2 Run A		Planck 2018 full likelihood	0.1583 ± 0.0033 (0.2σ from A.P.)
Level 2 Run D		Planck 2018 + RSD	0.1583 ± 0.0033 (0.2σ from A.P.)
All 17 chains		All combinations, $R-1 < 0.01$	Consistent at $< 0.3\sigma$, zero free params

The $\Delta\chi^2 = +0.54$ best-fit relative to CDM — with the Aristotelian Principle fixing the coupling and zero additional free parameters — indicates that the IAM framework is statistically indistinguishable from CDM under the full Planck likelihood while simultaneously predicting the observational tensions that CDM cannot address [Mahaffey,

5 The Cosmological Constant: A New Expression of the Principle

The most striking new application of the Aristotelian Principle is the relationship between the theoretical vacuum energy density predicted by quantum field theory and the observed cosmological constant. The 10^{122} -order-of-magnitude discrepancy between these quantities has been called the worst prediction in the history of science [Weinberg, 1989].

Within the IAM framework, this discrepancy is not a failed prediction. It is the expected ratio of accumulated actuality to total potential in a universe 13.8 billion years old. The vacuum energy ρ_{vac} represents the complete Planck-scale quantum substrate — the total unrealized potential of all field configurations available on the cosmic horizon. The cosmological constant Λ_{obs} represents the accumulated Landauer cost of baryonic gravitational decoherence events writing classical information onto that horizon since the electroweak transition. These are physically distinct quantities.

The Aristotelian Principle at cosmological scales gives:

$$\frac{\Lambda_{\text{obs}}}{\rho_{\text{vac}}} = \left(\frac{l_P}{l_H}\right)^2 \times \frac{\Omega_b}{\Omega_m} = 2.17 \times 10^{-123} \quad (2)$$

against the observed value of 1.14×10^{-123} — within a factor of 1.9 with no free parameters [Mahaffey, 2026f].

The Ω_b/Ω_m factor is not introduced to improve numerical agreement. It is a necessary consequence of IAM’s identification of dark matter as accumulated geometric potential — already classical, already actualized — which does not independently contribute to the ongoing information writing rate. Only baryons, as active decoherence engines on timelike worldlines, write new information onto the horizon. This identification was established in prior work [Mahaffey, 2026d] before the cosmological constant calculation was performed.

Equation (2) closes 121.9 of the 122 orders of magnitude of the cosmological constant problem with a calculation involving only the Planck length, the Hubble constant, and standard cosmological density parameters. The remaining factor of 1.9 is identified as the expected correction from the full cosmic decoherence history integral, which is the subject of ongoing work.

A secondary consequence: the cosmological constant is not strictly constant in IAM but increases slowly as $E(a)$ advances, predicting $w > -1$ — qualitatively consistent with DESI DR2 [DESI Collaboration, 2025].

This result extends the domain of the Aristotelian Principle beyond the $1/r$ potential systems that the virial theorem describes. The ratio $\Lambda/\rho_{\text{vac}}$ is not a virial ratio. It is the

fraction of Planck-scale potential that has been actualized through baryonic decoherence — the Aristotelian Principle operating at the ratio of the universe’s entire accumulated history to its total potential. The virial theorem is not being applied outside its domain. The Aristotelian Principle is being identified in a new domain where the virial theorem does not apply.

6 The General Statement of the Aristotelian Principle

Having surveyed the evidence across all physical domains, we state the Aristotelian Principle precisely.

The Aristotelian Principle: In any physical system where stable existence is maintained through the balance between unrealized potential and accumulated actuality, the ratio of the two is governed by the dimensionality of the space in which the system exists and the character of the potential that binds it. For systems governed by $1/r$ potentials in three-dimensional space, this ratio is exactly $1/2$ — the virial theorem. For systems in which the relevant potential is the complete Planck-scale vacuum energy and the relevant actuality is the accumulated Landauer cost of baryonic decoherence on the cosmic horizon, this ratio is $(l_P/l_H)^2 \times (\Omega_b/\Omega_m)$. Both are expressions of the same underlying condition: the equilibrium between what is unrealized and what has been actualized that is the condition of stable physical existence at every scale.

The virial theorem is the special case of the Aristotelian Principle for systems governed by $1/r$ potentials in three-dimensional space. It retains its name and its status as a precisely stated mathematical result within classical mechanics and statistical physics. The Aristotelian Principle names the broader physical law of which the virial theorem is one expression, and includes the cosmological applications, the information-theoretic applications, and any future applications at scales or in physical contexts where the $1/r$ potential does not apply but the potential-actual equilibrium condition does.

7 Petition for Naming

The virial theorem was named by Clausius in 1870 after the Latin *virial* (from *vis*, force or energy), reflecting the mathematical quantities it relates rather than the physical principle it expresses. Clausius found the special case — the $1/2$ partition for bound $1/r$ systems — and named it for what it uses mathematically.

The broader principle identified in this paper was identified philosophically before physics existed as a discipline. We propose naming it the **Aristotelian Principle** in recognition of this prior identification.

The justification proceeds in three parts.

First, the principle was identified philosophically 2,200 years before the virial theorem was written down. The physical and mathematical content of the principle — the identification of a universal equilibrium condition between unrealized potential and accumulated actuality as the condition of stable existence — is the central insight of a philosophical framework developed in the fourth century BC that identified the balance between these two aspects of existence as the fundamental condition of any persistent physical form.

Second, the naming honors the identification rather than the mathematical formulation. Scientific naming conventions frequently recognize the person who first identified a physical phenomenon, even when the precise mathematical statement came later. The naming of the Aristotelian Principle follows this convention: the principle was identified first, the mathematical expression came centuries later, and both deserve recognition.

Third, the naming is precise rather than metaphorical. The Aristotelian Principle as stated in Section 6 is the mathematical realization of the same physical insight in the language of modern physics. The identification is not that physics resembles ancient philosophy. The identification is that ancient philosophy was describing a physical law that has now been given mathematical form and verified across 22 physical domains from the hydrogen atom to the cosmological constant.

The virial theorem retains its name as the special case for bound $1/r$ systems. The Aristotelian Principle names the general law. The two names coexist, with the virial theorem as the historically established special case and the Aristotelian Principle as the newly identified general principle.

8 Conclusions

We have presented evidence for the Aristotelian Principle as a universal equilibrium condition operating across every physical domain in which the IAM framework has been tested. Tables 1 and 2 document 22 expressions of the principle across 33 papers, from the hydrogen atom to the cosmological constant. The evidence includes:

- Machine-precision verification in electromagnetic systems (atoms and molecules) — the virial theorem at its most exact.
- Exact observational confirmation at the Chandrasekhar white dwarf limit — the virial theorem defining a physical quantity.

- MCMC confirmation of the cosmological coupling constant $\beta_m = \Omega_m/2$ at 0.2σ from 17 independent Planck 2018 chains — the strongest empirical test of the principle.
- The cosmological constant ratio $\Lambda/\rho_{\text{vac}} = (l_P/l_H)^2 \times (\Omega_b/\Omega_m) = 2.17 \times 10^{-123}$ within a factor of 1.9 of the observed value with no free parameters — the Aristotelian Principle beyond the virial theorem’s domain.
- The structural exception of the weak nuclear force — confirming that the principle distinguishes forces of stable structure from the force of irreversible change.

The virial theorem is the special case of the Aristotelian Principle for bound $1/r$ systems. Clausius found the special case in 1870 and named it for the mathematics. The general principle operates at every scale from the hydrogen atom to the cosmic horizon, across electromagnetic, gravitational, and information-theoretic domains. We propose naming it in honour of its original identification.

The physical content of the Aristotelian Principle — the equilibrium between what has not yet been actualized and what has been irreversibly actualized — runs through every result in the IAM framework and connects them into a single coherent web. This is not an application of the virial theorem to domains where it does not apply. It is the identification of the deeper principle of which the virial theorem is one expression.

It is perhaps worth noting, as a matter of historical observation, that the equilibrium condition between unrealized potential and accumulated actuality — described mathematically in this paper as governing physical systems from the hydrogen atom to the cosmic horizon — was identified as the fundamental condition of finite existence by a natural philosopher 2,200 years before Clausius wrote the first equation. Whether this represents a remarkable coincidence of intellectual history or something more deeply connected to the structure of the principle itself is a question the authors leave to the reader.

Data Availability

All MCMC chains, analysis scripts, and computational notebooks are publicly available at <https://github.com/hmahaffeyges/IAM-Validation> (Zenodo DOI: 10.5281/zenodo.18702042).

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